ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 8

MATHEMATICS

PAPER 1 · PRACTICE D

MARKS: 100 **TIME: 2 hours**

Covers the full Grade 8 year: profit and loss, compound interest, fuel cost and exchange rates, squares, cubes and roots, scientific notation, integers and exponent laws, fractions in context, input–output tables, substitution, expressions and equations, vertically opposite, alternate and co-interior angles, similar triangles, quadrilaterals and angles around a point, Pythagoras, triangular prisms, circles and unit conversions, data from a list of marks, and a translation on the Cartesian plane.

Written by Academic Central for practice, following the CAPS curriculum for Grade 8. It is not a Department of Basic Education examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of NINE questions. Answer ALL the questions.
2. Number your answers exactly as the questions are numbered.
3. Show ALL your working. Answers alone will not get full marks.
4. A non-programmable calculator may be used, unless a question says otherwise.
5. Round answers to TWO decimal places where necessary. Diagrams are NOT drawn to scale.
6. Give reasons for your statements in geometry questions.

QUESTION 1: MULTIPLE CHOICE

Choose the correct answer. Write only the letter (A–D) next to the question number, e.g. 1.11 B.

1.1 $\sqrt{144} + \sqrt[3]{27} = \ldots$

A 12 B 15 C 18 D 39

1.2 $4{,}5 \times 10^{-3}$ written as an ordinary number is …

A 4 500 B 0,045 C 0,45 D 0,004 5

1.3 $-8 - (-12) = \ldots$

A 4 B −20 C −4 D 20

1.4 The highest common factor (HCF) of 18 and 24 is …

A 3 B 72 C 6 D 12

1.5 $3a^2b \times 4ab^3 = \ldots$

A $12a^3b^4$ B $7a^3b^4$
C $12a^2b^3$ D $7a^2b^3$

1.6 Which of these angles is a reflex angle?

A 90° B 179° C 45° D 200°

1.7 Co-interior angles between two parallel lines add up to …

A 90°. B 180°. C 360°. D 270°.

1.8 The point (3 ; −2) is reflected in the *y*-axis. The image is …

A (3 ; 2) B (−3 ; 2) C (−3 ; −2) D (2 ; −3)

1.9 A fair coin is tossed twice. The probability of getting two heads is …

A $\frac{1}{2}$ B $\frac{3}{4}$ C $\frac{1}{3}$ D $\frac{1}{4}$

1.10 The median of 3 ; 9 ; 4 ; 7 ; 5 is …

A 5 B 7 C 4 D 6

[10]

QUESTION 2: FINANCE

2.1 Thabo buys 40 packets of sweets at R2,50 each and sells all of them at R4 each.

2.1.1 Calculate how much he paid for the sweets. (1)

2.1.2 Calculate his profit. (1)

2.1.3 Calculate his profit as a percentage of the cost price. (2)

2.2 Naledi invests R5 000 at 8% compound interest per year. Calculate how much she will have after 2 years. (3)

2.3 A car uses 7 litres of petrol for every 100 km. Petrol costs R23,50 per litre. Calculate the cost of petrol for a trip of 450 km. (3)

2.4 The exchange rate is US$1 = R18,20. How many US dollars will you get for R1 000? (2)

[12]

QUESTION 3: EXPONENTS AND INTEGERS

3.1 Simplify: (a) $(2a^3)^2 \times 3a^0$ (b) $(x^5 \times x^3) \div x^4$ (c) $\sqrt{(16x^8)}$ (6)

3.2 Write in scientific notation: (a) 3 450 000 (b) 0,000 062 (2)

3.3 A diver is 18 m below sea level. She rises 7 m and then dives down another 12 m. What is her position now? (2)

[10]

QUESTION 4: FRACTIONS AND DECIMALS

4.1 Calculate WITHOUT a calculator: (a) $2\frac{1}{3} - 1\frac{3}{4}$ (b) $\frac{3}{5} \div \frac{9}{10}$ (c) 1,2 × 0,05 (5)

4.2 A school has 120 Grade 8 learners. $\frac{3}{8}$ of them walk to school, 0,25 of them come by taxi and the rest come by bus. How many learners come by bus? (3)

[8]

QUESTION 5: PATTERNS, FUNCTIONS AND ALGEBRA

5.1 Study the input–output table.

x	1	2	3	4	10	*b*
y	5	8	11	14	*a*	62

5.1.1 Write down the rule for *y* in terms of *x*. (2)

5.1.2 Calculate *a*. (1)

5.1.3 Calculate *b*. (1)

5.2 Simplify: (a) $4x - 3(x - 5)$ (b) $(6a^2b - 9ab^2) \div 3ab$ (4)

5.3 If $a = -2$ and $b = 3$, calculate the value of $a^2 - 2ab + b^3$. (3)

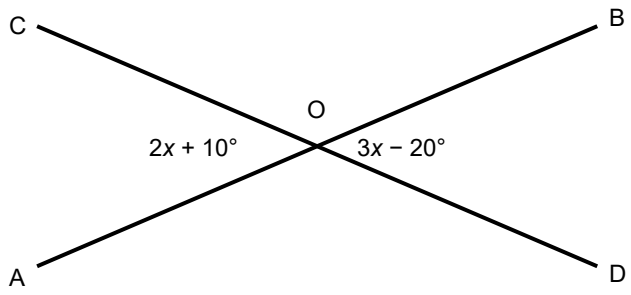
5.4 Solve for *x*: (a) $5x - 7 = 2x + 8$ (b) $3(x - 4) = -6$ (c) $\frac{x}{4} - 1 = 2$ (6)

5.5 The perimeter of a rectangle is 46 cm. Its length is 5 cm more than its width. Write an equation and solve it to find the length and the width. (3)

[20]

QUESTION 6: GEOMETRY

6.1 Straight lines AB and CD intersect at O. ∠AOC = $2x + 10°$ and ∠BOD = $3x - 20°$.



Calculate, giving reasons: (a) the value of x (b) ∠AOD (4)

6.2 KL ∥ MN. A straight line cuts KL at P and MN at Q, with K and M on the same side of PQ. ∠KPQ = 65°. Calculate, giving reasons: (a) ∠PQN (b) ∠PQM (4)

6.3 ΔABC is similar to ΔDEF (ΔABC ||| ΔDEF). AB = 4 cm, BC = 6 cm and DE = 10 cm. Calculate the length of EF. (2)

6.4 Name the quadrilateral: (a) all four sides are equal and all four angles are 90°; (b) it has only one pair of parallel sides; (c) all four sides are equal, the diagonals bisect each other at 90°, but the angles are not 90°. (3)

6.5 Three angles around a point are x, $2x$ and 150°. Calculate x. Give a reason. (3)

[16]

QUESTION 7: PYTHAGORAS AND MEASUREMENT

7.1 A 5 m ladder leans against a vertical wall. The foot of the ladder is 1,4 m from the wall. Calculate how high up the wall the ladder reaches. (3)

7.2 A triangular prism has a triangular end with a base of 6 cm and a perpendicular height of 4 cm. The prism is 10 cm long. Calculate its volume. (3)

7.3 A circle has a diameter of 14 cm. Use $\pi = {}^{22}\!/_{7}$ to calculate (a) its circumference and (b) its area. (4)

7.4 Convert: (a) 2,5 m² to cm² (b) 3 500 cm³ to litres (4)

[14]

QUESTION 8: DATA HANDLING

8.1 The test marks (out of 100) of 12 learners are: 45 ; 52 ; 60 ; 60 ; 67 ; 71 ; 74 ; 74 ; 74 ; 80 ; 85 ; 98

8.1.1 Write down the mode. (1)

8.1.2 Calculate the median. (2)

8.1.3 Calculate the range. (1)

8.1.4 Calculate the mean. (2)

[6]

QUESTION 9: TRANSFORMATIONS

9.1 Triangle ABC has vertices A(1 ; 2), B(4 ; 2) and C(1 ; 5). It is translated 3 units to the left and 2 units down.

9.1.1 Write down the coordinates of A′, B′ and C′. (3)

9.1.2 Is ΔA′B′C′ congruent to ΔABC? Give a reason. (1)

[4]

TOTAL: 100

END OF QUESTION PAPER